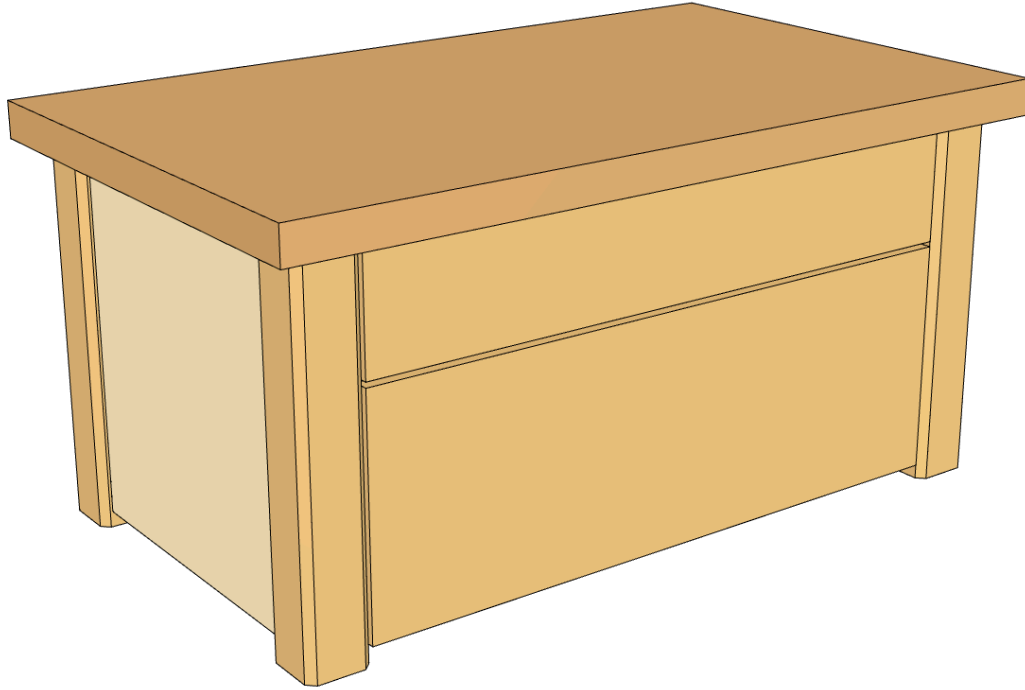




DRAWER BENCH · 40 X 24 X 19-5/8 IN

Drawer Bench Cut List

25 pieces: the case first, then the drawers, each grouped by material in cutting order. Inches to the nearest 1/32. Every part has a shop drawing with its grooves, rabbets, notches and bores located from the blank's own edges; tap a drawing to zoom. Tap a row to tick it off; ticks stay on this phone only. The assembly order with a picture per step is at the end.

Provisional. Every groove, rabbet and dado that takes plywood is sized to the sheet's real thickness and says "measure". Measure the sheets you bought, put the numbers in the model and re-run it before cutting joinery. The top's overhang and edge profile still wait on the island; the slide numbers wait on the purchased sheet.

Case 13 pieces

Soft maple: posts, rails, cleat

8 pieces · 2.6 sq ft face, no kerf or waste

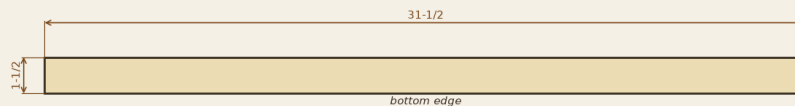
1× cleat_rear

3/4 x 1-1/2 x 31-1/2

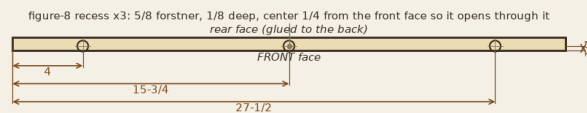
REAR top cleat, 3/4 rail stock x 1-1/2 tall, between the posts; glued to the inside face of the back and clamped with #8 x 1-1/4 screws countersunk from the cleat's inside face (nothing through the back), top edge flush with the post tops; three figure-8 fasteners on top: 5/8 dia x 1/8 deep forstner recesses centered 1/4 from the front (inner) face, so each opens through it and the fastener hangs over into the cabinet, at 4, 15-3/4 and 27-1/2 from either end (blank datums), #8 x 5/8 screws into the cleat and into the top (replaces the tenoned rear rail)

cleat_rear x1 | REAR top cleat, 3/4 soft maple, 1-1/2 tall x 31-1/2 long

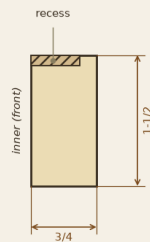
face view



top edge seen from above; FRONT face (inside the cabinet) along the bottom



end view; inner face along the left

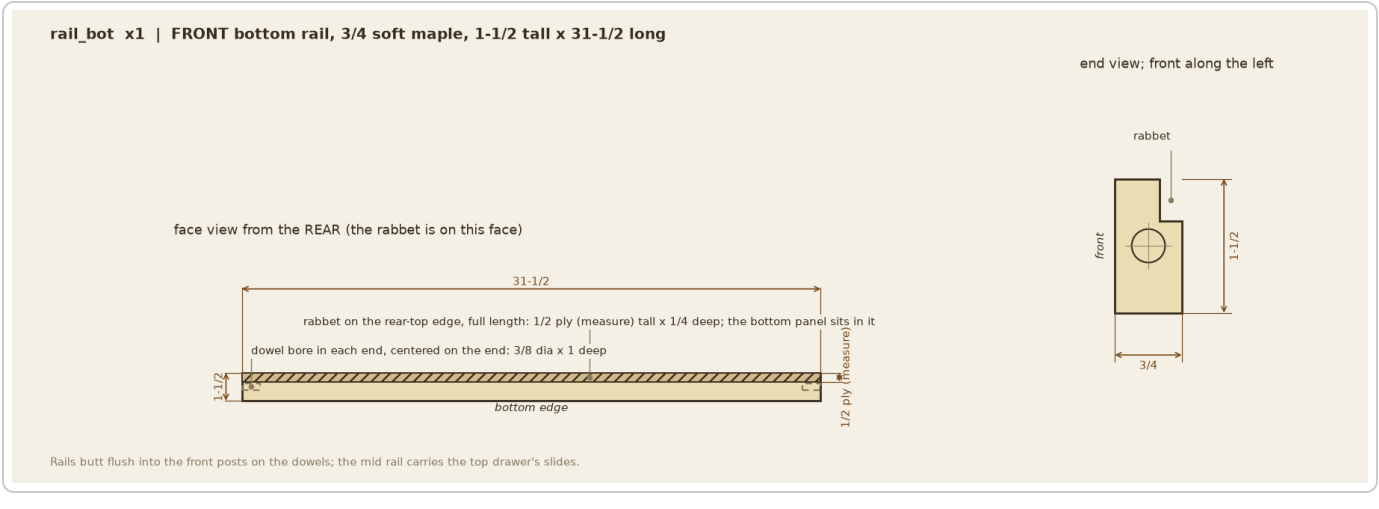


Glued to the inside face of the back, top edge flush with the post tops; clamp with #8 x 1-1/4 screws from this face.

1x rail_bot see drawing: not a plain rectangle

3/4 x 1-1/2 x 31-1/2

FRONT bottom rail, 3/4 stock: butts flush into the post faces, 3/8 dia x 1 deep dowel each end; rabbet 1/4 deep x 1/2 ply (measure) tall on the rear-top edge, full length, for the bottom panel; glue and screw the bottom into it (no lip above)



1x rail_top

3/4 x 1 x 31-1/2

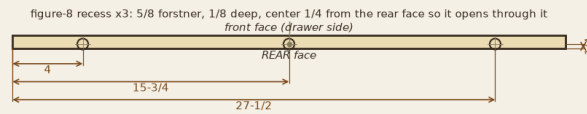
FRONT top rail, 3/4 stock: butts flush into the post faces, 3/8 dia x 1 deep dowel each end; three figure-8 fasteners on top: 5/8 dia x 1/8 deep forstner recesses centered 1/4 from the rear (inner) face, so each opens through it and the fastener hangs over into the cabinet, at 4, 15-3/4 and 27-1/2 from either end (blank datums), #8 x 5/8 screws into the rail and into the top

rail_top x1 | FRONT top rail, 3/4 soft maple, 1 tall x 31-1/2 long

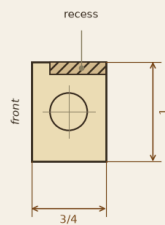
face view



top edge seen from above; REAR face (inside the cabinet) along the bottom



end view; front along the left



Rails butt flush into the front posts on the dowels; the mid rail carries the top drawer's slides.

1× rail_mid

3/4 x 1 x 31-1/2

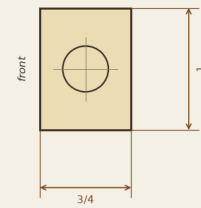
FRONT mid rail, 3/4 stock: butts flush into the post faces, 3/8 dia x 1 deep dowel each end; carries the top drawer slides

rail_mid x1 | FRONT mid rail, 3/4 soft maple, 1 tall x 31-1/2 long

face view



end view; front along the left

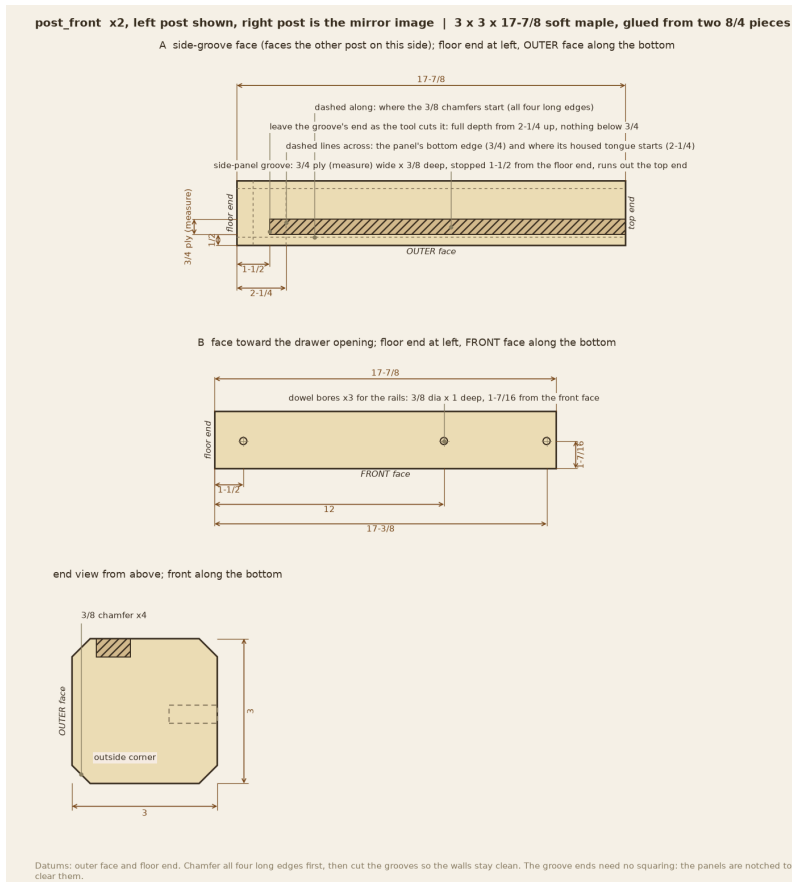


Rails butt flush into the front posts on the dowels; the mid rail carries the top drawer's slides.

2x post_front see drawing: not a plain rectangle

3 x 3 x 17-7/8

one of 2 FRONT posts, left/right mirrored, grooves on the inner faces; glue-up of two 8/4 pieces milled to 1-1/2; 3/8 chamfer x4 edges full length; side groove 3/4 ply (measure) wide x 3/8 deep at 1/2 from the outer face, stopped 1-1/2 above the floor, open at top; leave each groove's end as the tool cuts it (rounded or ramped is fine): the panel's housed tongue starts 2-1/4 above the floor, 3/4 above the stop, and the panel's notched bottom covers the end; nothing below 3/4 (the panel's bottom edge); grain vertical, glue-up seam on a side face; three 3/8 dia x 1 deep dowel bores on the inner face 1 in behind the front face, centered on the rail joints 1-1/2, 12 and 17-3/8 above the floor



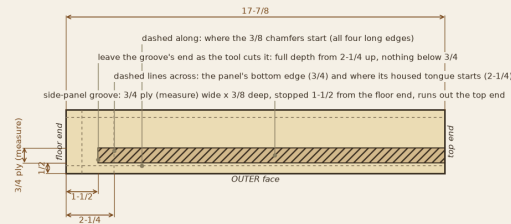
2x post_rear see drawing: not a plain rectangle

3 x 3 x 17-7/8

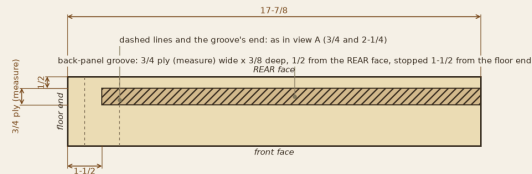
one of 2 REAR posts, left/right mirrored, grooves on the inner faces; same blank and chamfer as post_front; side groove as post_front; back groove 3/4 ply (measure) wide x 3/8 deep on the inner face at 1/2 from the rear face, stopped 1-1/2 above the floor, open at top; no mortises; leave each groove's end as the tool cuts it (rounded or ramped is fine): the panel's housed tongue starts 2-1/4 above the floor, 3/4 above the stop, and the panel's notched bottom covers the end; nothing below 3/4 (the panel's bottom edge); grain vertical

post_rear x2, left post shown, right post is the mirror image | 3 x 3 x 17-7/8 soft maple, glued from two 8/4 pieces

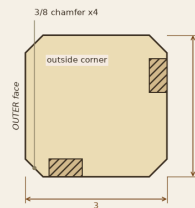
A side-groove face (faces the other post on this side); floor end at left, OUTER face along the bottom



B face toward the other rear post; floor end at left, REAR face along the top



end view from above; front along the bottom



Datums: outer face and floor end. Chamfer all four long edges first, then cut the grooves so the walls stay clean. The groove ends need no squaring: the panels are notched to clear them.

3/4 in plywood (measure the sheet): sides and back

3 pieces · 7.7 sq ft face, no kerf or waste

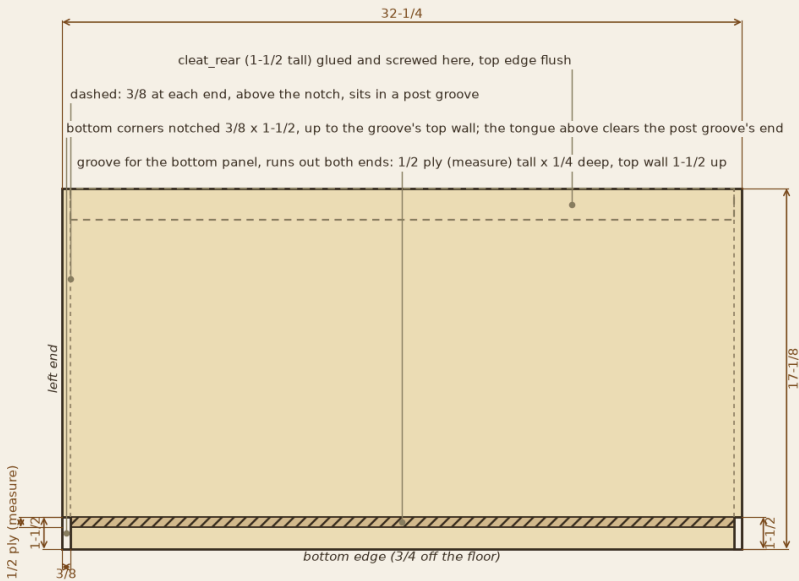
1x back

23/32 x 17-1/8 x 32-1/4

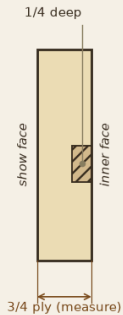
housed 3/8 in each post above the notches; full height, top edge flush with the post tops; cleat_rear glued and screwed inside along the top edge; both bottom corners notched 3/8 (from each end) x 1-1/2 tall: the housed tongue starts 3/4 above the post groove's end and the panel's bottom 1-1/2 butts the post face; through groove 1/4 deep x 1/2 ply (measure) tall on the inner face for the bottom, full length, top wall 1-1/2 above the bottom edge

back x1 | 3/4 ply (measure), 32-1/4 long x 17-1/8 tall, face grain vertical

inner face (toward the cabinet); bottom edge along the bottom



section, bottom edge



The groove runs out both ends. Above the notches the ends sit 3/8 deep in the post grooves; below them the panel butts the post face and covers the groove's end, so nothing shows.

23/32 x 16-1/4 x 17-1/8

The groove runs out both ends. Above the notches the ends sit 3/8 deep in the post grooves; below them the panel butts the post face and covers the groove's end, so nothing shows.

1/2 in plywood (measure the sheet): case bottom

1 pieces · 4.7 sq ft face, no kerf or waste

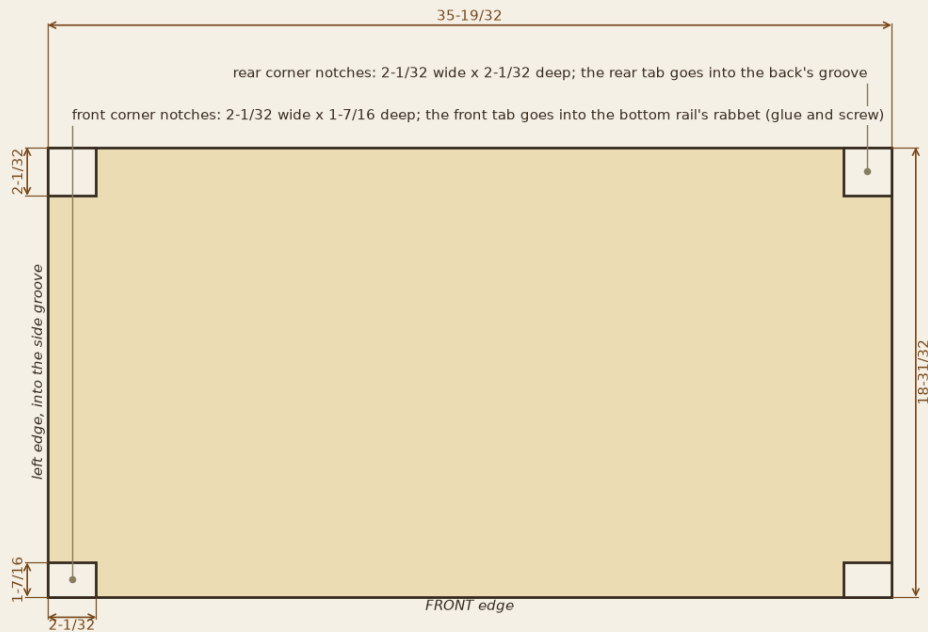
1× bottom see drawing: not a plain rectangle

15/32 x 18-31/32 x 35-19/32

notch the four corners 2-1/32 wide x 1-7/16 (front) / 2-1/32 (rear) for the posts; edges in the side/back grooves and the rail_bot rabbet (glue and screw the front edge)

bottom x1 | 1/2 ply (measure), 35-19/32 x 18-31/32 overall, four corners notched around the posts

plan view from above; FRONT edge along the bottom



Notch datums are the blank's edges. Pre-join to the back off the bench (see design.md, assembly order).

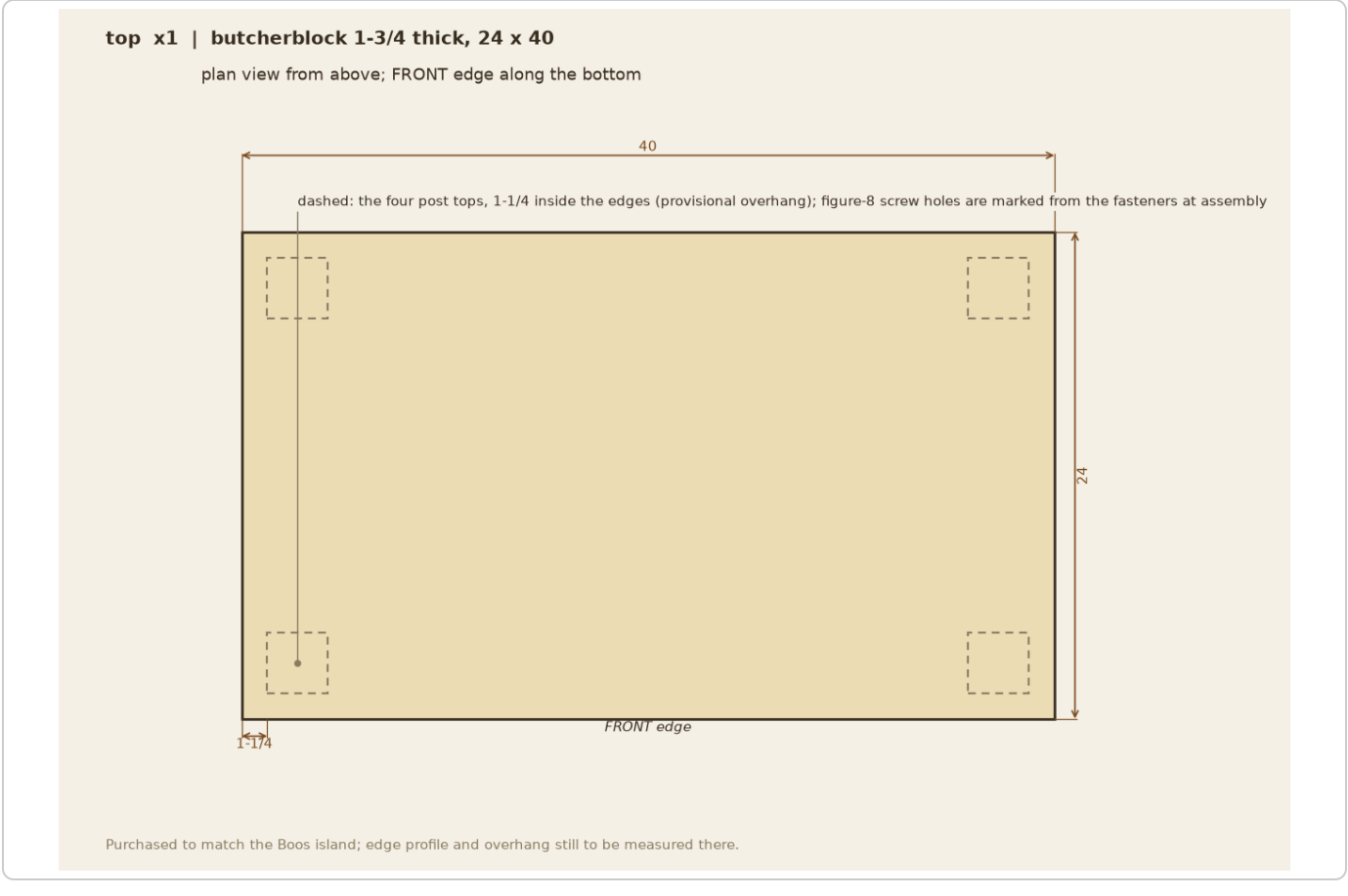
Butcherblock top (purchased, Boos match)

1 pieces · 6.7 sq ft face, no kerf or waste

1x top

1-3/4 x 24 x 40

provisional 1-3/4 thick, 1-1/4 overhang all round; edge profile to match the island; six figure-8 fasteners (three into rail_top, three into cleat_rear, #8 x 5/8 screws both ends), no glue



Drawers 12 pieces

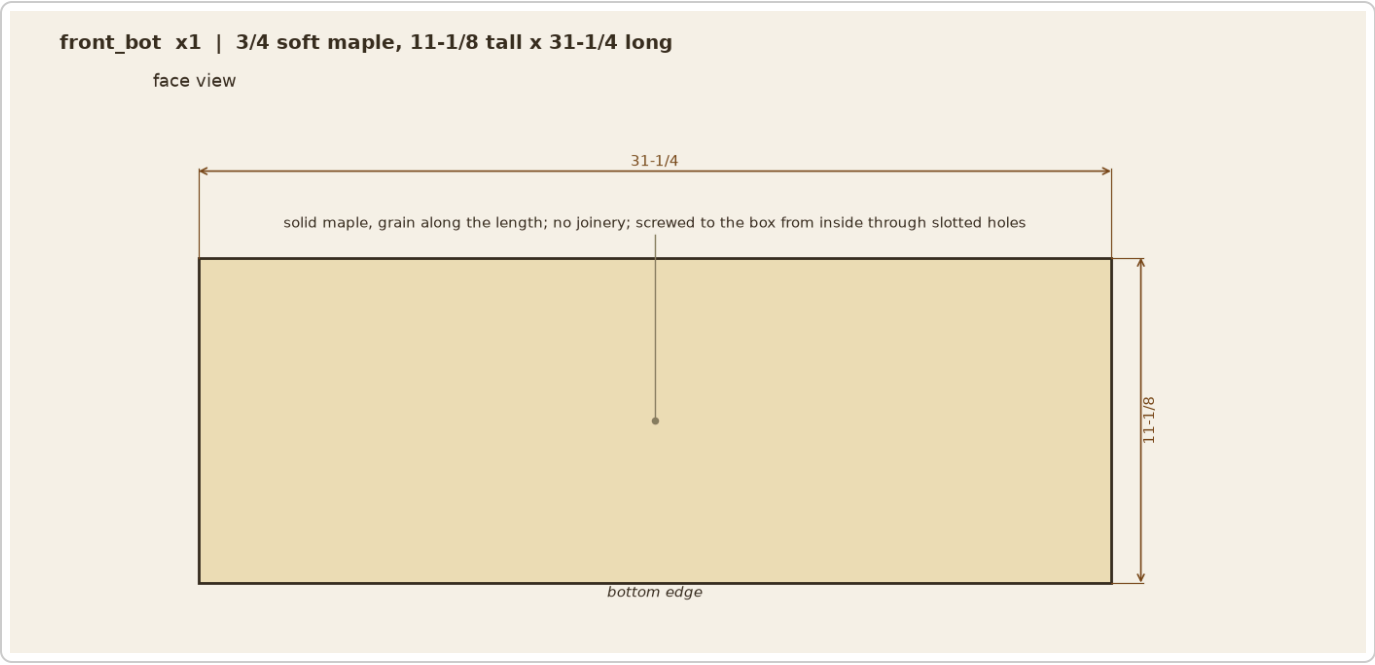
Drawer fronts, soft maple

2 pieces · 3.6 sq ft face, no kerf or waste

1x front_bot

3/4 x 11-1/8 x 31-1/4

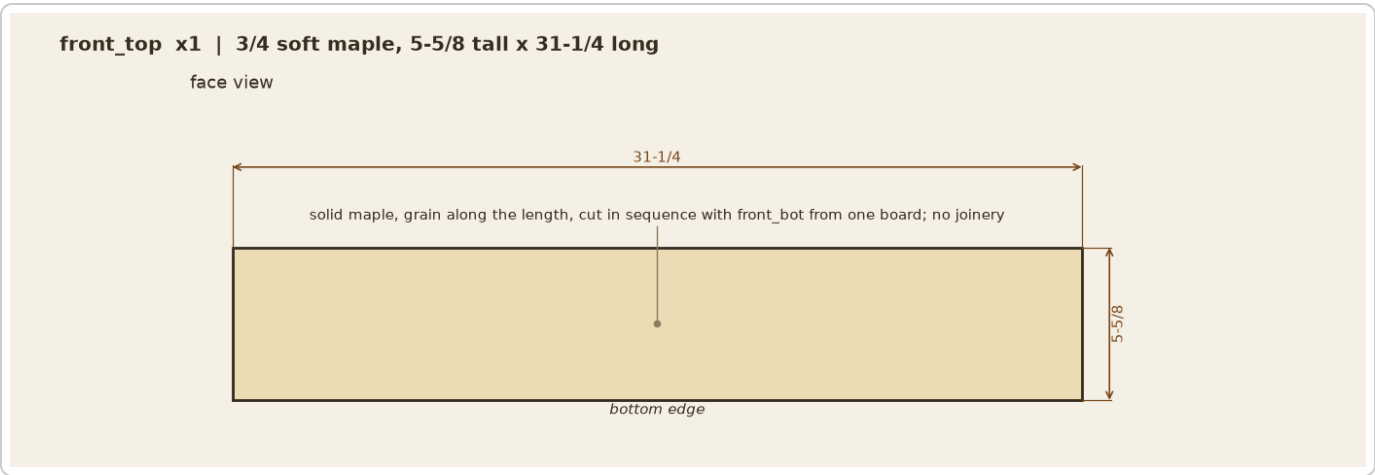
grain along the length; screwed to the box from inside through slotted holes (cross-grain 11-1/8 wide); pull undecided; leave about 1/16 between the back face and the rails (Blum front gap)



1x front_top

3/4 x 5-5/8 x 31-1/4

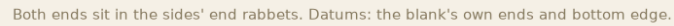
grain along the length; screwed to the box from inside; pull undecided; leave about 1/16 between the back face and the rails (Blum front gap), the runner setback sets the closed position



8 pieces · 8.1 sq ft face, no kerf or waste

$$15/32 \times 8 - 15/32 \times 30 - 11/32$$

inner face (toward the inside of the box)



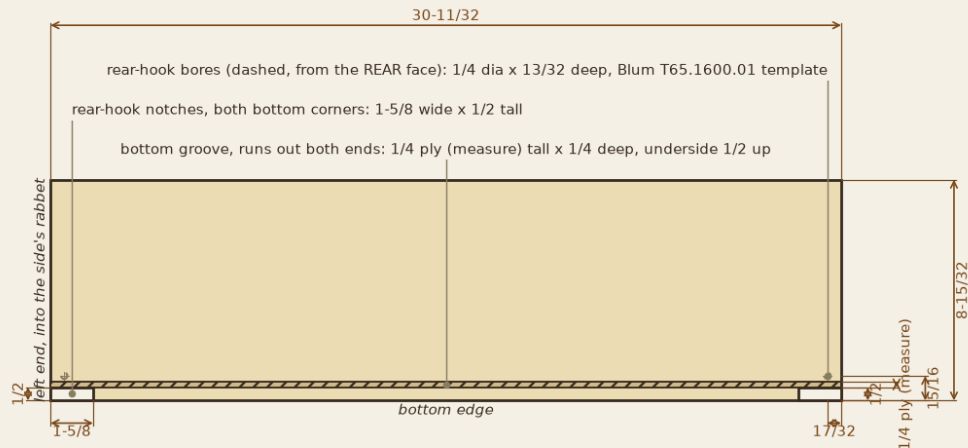
1x drawer_back_bot see drawing: not a plain rectangle

15/32 x 8-15/32 x 30-11/32

box back, datums from the blank's ends: bottom groove 1/4 deep x 1/4 ply (measure) tall with its underside 1/2 above the bottom edge, runs out both ends; rear-hook notches 1-5/8 from each end x 1/2 tall at both bottom corners; rear-hook bores 1/4 dia x 13/32 deep from the rear face, centered 17/32 from each end (= 9/32 from the side's inner face) and 15/16 above the bottom edge (Blum T65.1600.01 template)

drawer_back_bot x1 | 1/2 ply (measure), 8-15/32 tall x 30-11/32 long (box back)

inner face (toward the inside of the box)



Both ends sit in the sides' end rabbets. Datums: the blank's own ends and bottom edge.

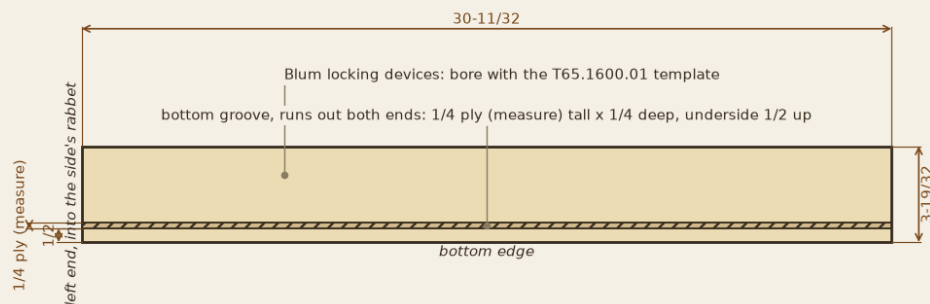
1x drawer_front_top see drawing: not a plain rectangle

15/32 x 3-19/32 x 30-11/32

box front (sub-front): bottom groove 1/4 deep x 1/4 ply (measure) tall with its underside 1/2 above the bottom edge; locking devices bored with the Blum T65.1600.01 template

drawer_front_top x1 | 1/2 ply (measure), 3-19/32 tall x 30-11/32 long (box front, behind the maple front)

inner face (toward the inside of the box)



Both ends sit in the sides' end rabbets. Datums: the blank's own ends and bottom edge.

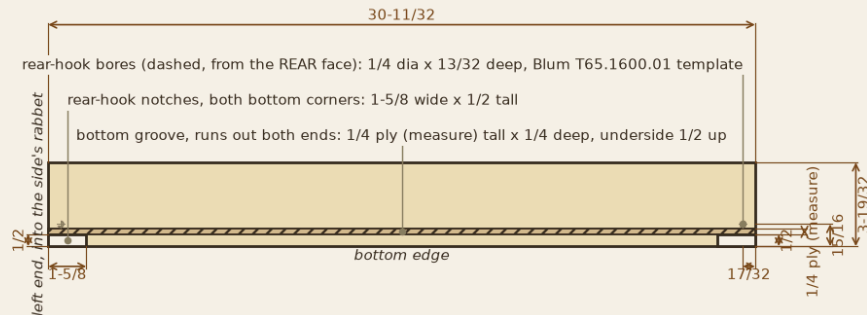
1× drawer_back_top see drawing: not a plain rectangle

15/32 x 3-19/32 x 30-11/32

box back, datums from the blank's ends: bottom groove 1/4 deep x 1/4 ply (measure) tall with its underside 1/2 above the bottom edge, runs out both ends; rear-hook notches 1-5/8 from each end x 1/2 tall at both bottom corners; rear-hook bores 1/4 dia x 13/32 deep from the rear face, centered 17/32 from each end (= 9/32 from the side's inner face) and 15/16 above the bottom edge (Blum T65.1600.01 template)

drawer_back_top x1 | 1/2 ply (measure), 3-19/32 tall x 30-11/32 long (box back)

inner face (toward the inside of the box)



Both ends sit in the sides' end rabbets. Datums: the blank's own ends and bottom edge.

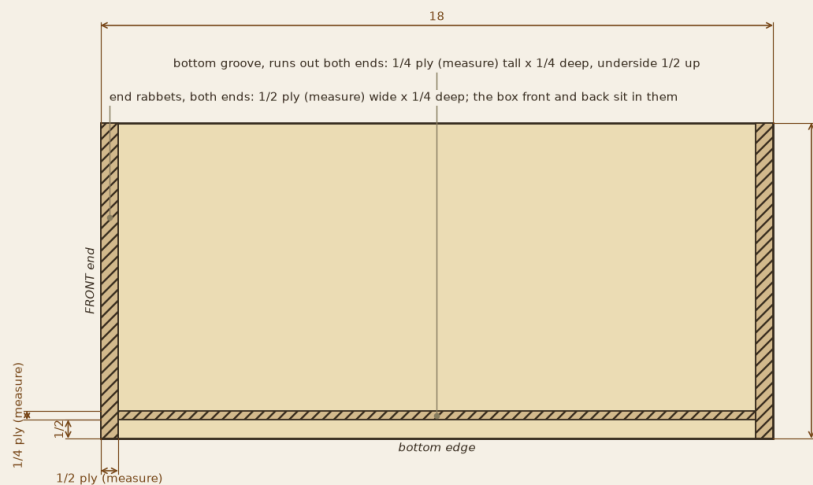
2× drawer_side_bot see drawing: not a plain rectangle

15/32 x 8-15/32 x 18

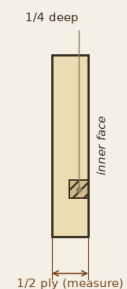
end rabbets 1/4 deep x 1/2 ply (measure) wide; bottom groove 1/4 deep x 1/4 ply (measure) tall with its underside 1/2 above the bottom edge

drawer_side_bot x2, left shown, right is the mirror image | 1/2 ply (measure), 8-15/32 tall x 18 long

inner face (toward the inside of the box); FRONT end at left



section, bottom edge



Undermount box: the bottom groove sits 1/2 in up so the Blum runner clears under the bottom.

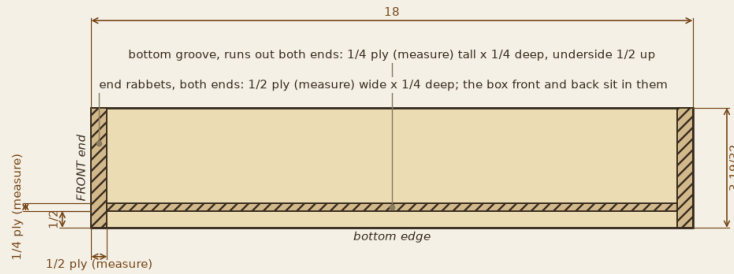
2× drawer_side_top see drawing: not a plain rectangle

15/32 x 3-19/32 x 18

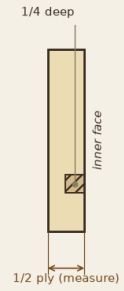
end rabbets 1/4 deep x 1/2 ply (measure) wide; bottom groove 1/4 deep x 1/4 ply (measure) tall with its underside 1/2 above the bottom edge

drawer_side_top x2, left shown, right is the mirror image | 1/2 ply (measure), 3-19/32 tall x 18 long

inner face (toward the inside of the box); FRONT end at left



section, bottom edge



Undermount box: the bottom groove sits 1/2 in up so the Blum runner clears under the bottom.

Drawer bottoms, 1/4 in plywood (measure the sheet)

2 pieces · 7.4 sq ft face, no kerf or waste

2× drawer_bottom_bot/drawer_bottom_top

1/4 x 17-9/16 x 30-11/32

plain rectangle; sits in the 1/4 deep grooves of the sides, front and back

drawer_bottom_bot x1 | 1/4 ply (measure), 17-9/16 x 30-11/32

plan view; FRONT edge along the bottom, REAR edge along the top



Sits in the 1/4 in grooves of the sides, front and back, 1/2 in above the box's bottom edges.

Hardware from the design; not in the cut list

2 pairs **Blum TANDEM plus BLUMOTION 563H4570B undermount slides, 18 in class**
confirm the purchased spec against the 563H sheet before boring the boxes

2 pairs **Blum rear mounting brackets 295.3750.02**
on the back panel, at runner height

2 sets **Blum locking devices for the box fronts**
bored with the T65.1600.01 template

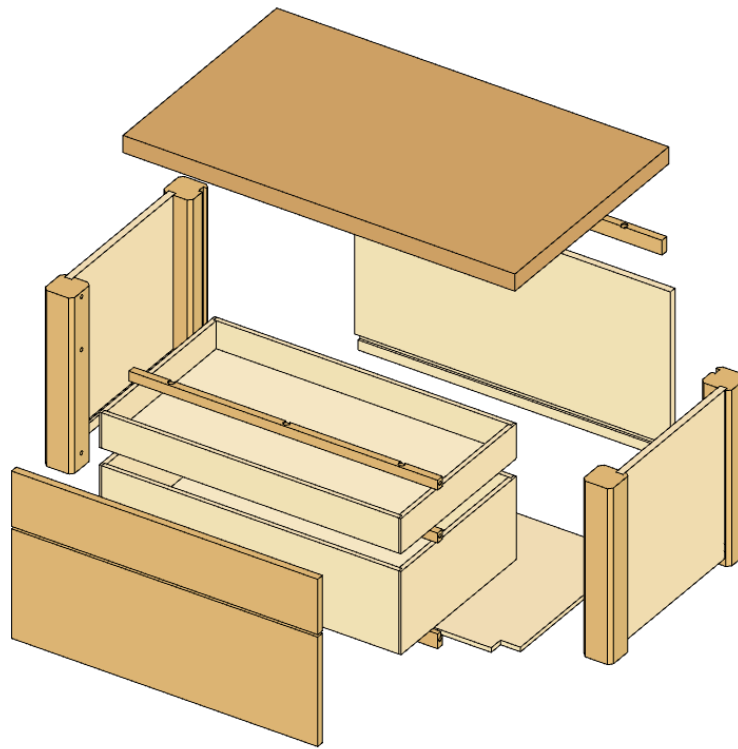
6 **3/8 in dowels, 2 in long**
front rails to front posts, one each end, 1 in into each part

6 **figure-8 desktop fasteners with #8 x 5/8 in screws (12)**
three on rail_top, three on cleat_rear, in the offset 5/8 in recesses

4 to 5 **#8 x 1-1/4 in screws, countersunk**
cleat_rear to the inside of the back, from the cleat side

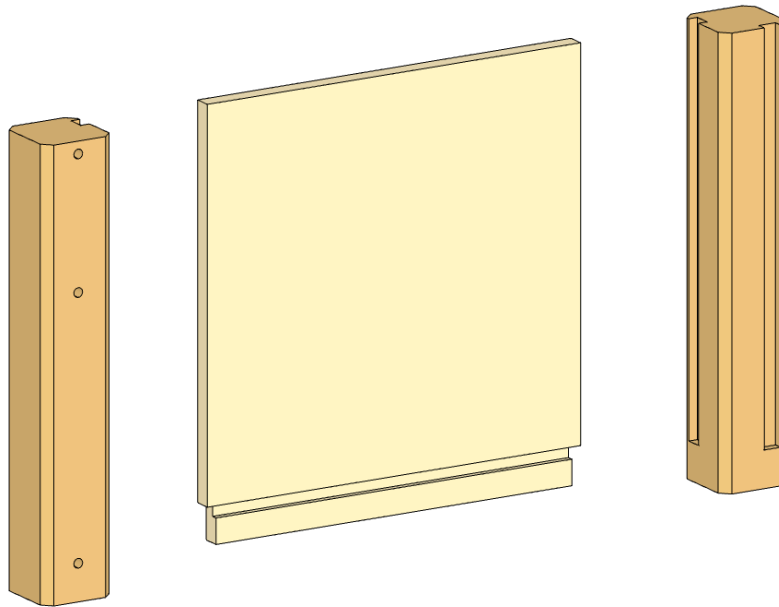
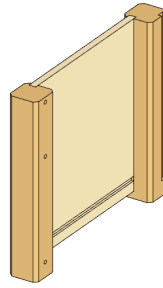
as **screws for the drawer fronts, through slotted holes from inside the boxes**
needed pulls undecided

Assembly 7 steps, in order



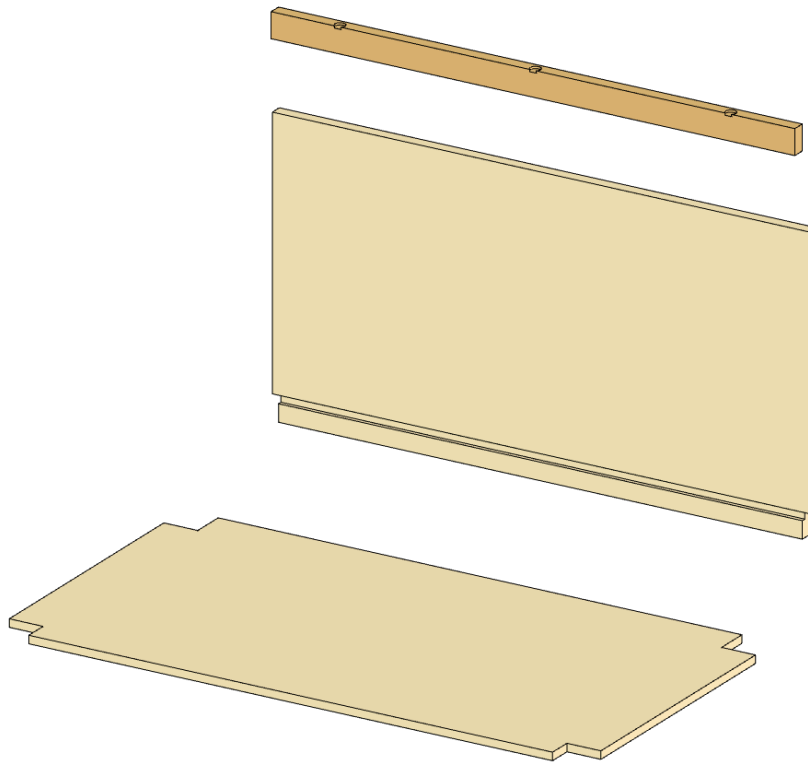
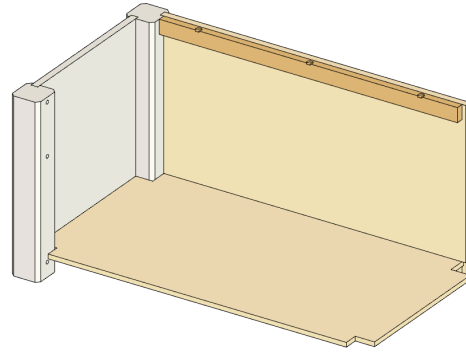
1. Side frames

Glue each side panel into its front and rear posts: the $\frac{3}{8}$ tongues above the notches go into the stopped grooves, glue on both faces of each tongue. Stand the posts and the panel on the bench with a $\frac{3}{4}$ block under the panel's bottom edge; that puts the panel's top edge flush with the post tops. Check flush with a straightedge before clamping. Make both sides now: the left one stands up first, the right one closes the case in step 4.



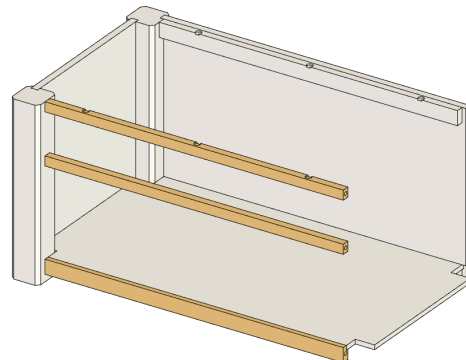
2. Back, cleat and bottom

Off the bench, glue the cleat inside the back's top edge, flush with the top, figure-8 recesses up, and clamp it with the #8 x 1-1/4 screws from the cleat side. Push the bottom's rear edge into the back's groove with glue. Stand the left side frame and slide the pair in from the right, both at once: the back's tongue into the rear post's groove, the bottom's side edge into the side panel's groove.



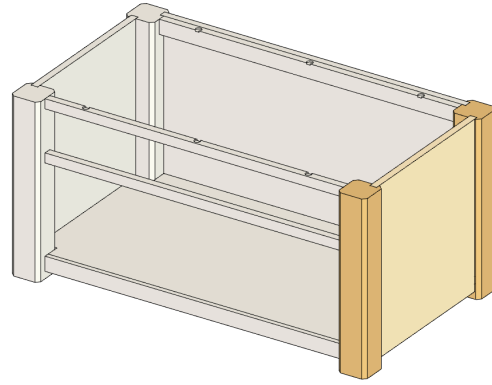
3. Front rails

Dowel the three rails into the left front post, two 3/8 dowels each, glue. The bottom's front edge sits in the bottom rail's rabbet, flush with the rail's top: glue it and screw down through the bottom into the rail. The top rail's figure-8 recesses face up and open toward the inside.



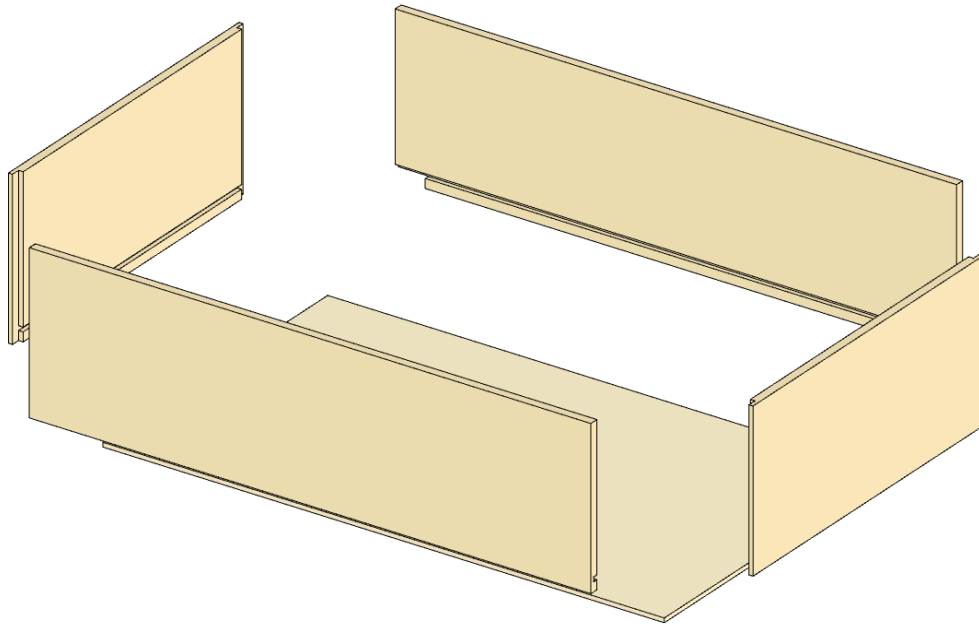
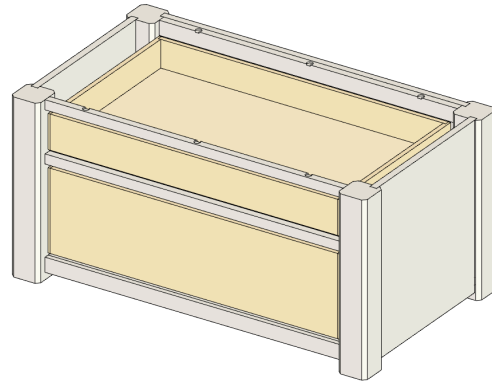
4. Close the right side

Bring the right side frame in over the back's tongue, the bottom's edge and the three rails' dowels at once, all glued. Clamp across the case side to side at the posts, check the front frame square and the post tops level, and leave it until the glue cures.



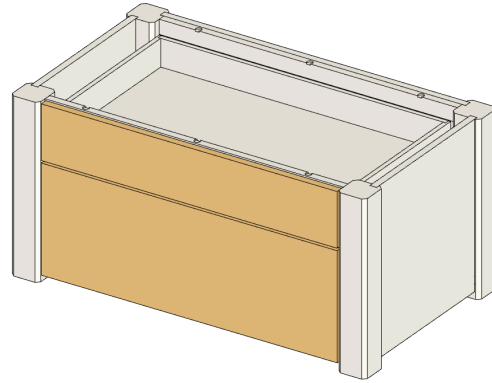
5. Slides and drawer boxes

Glue each box up around its bottom: front and back in the sides' end rabbets, the 1/4 bottom in the grooves with its underside 1/2 above the sides' bottom edges, hook notches and bores at the rear. Screw the Blum runners to the posts' inner faces and the rear brackets to the back, clip the locking devices under the box fronts and set the boxes on the runners.



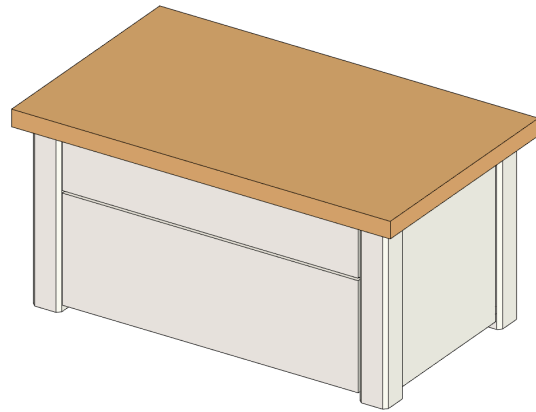
6. Drawer fronts

Screw the fronts to the box fronts from inside through oversize holes: 1/8 to the posts, 1/8 under the top, 1/4 between the two, the bottom one 3/4 off the floor. Pulls once chosen.



7. Top

Set the top with 1-1/4 overhang all round and fix it with the six figure-8 fasteners on the top rail and the cleat, #8 x 5/8 screws. No glue; the fasteners let it move.



Generated 2026-10-07 from `projects/Drawer-bench/cutlist.csv` and `images/parts/`, which `drawer_bench.py` and `part_drawings.py` write on every export; every drawn cut is probed against the CAD solid before its sheet is written. The bench image at the top and the assembly pictures are screenshots of the CAD viewer (`hero_shot.py`, `assembly_shots.py`); the PDF is this page printed by `make_pdf.py`. Part names match the CAD viewer.